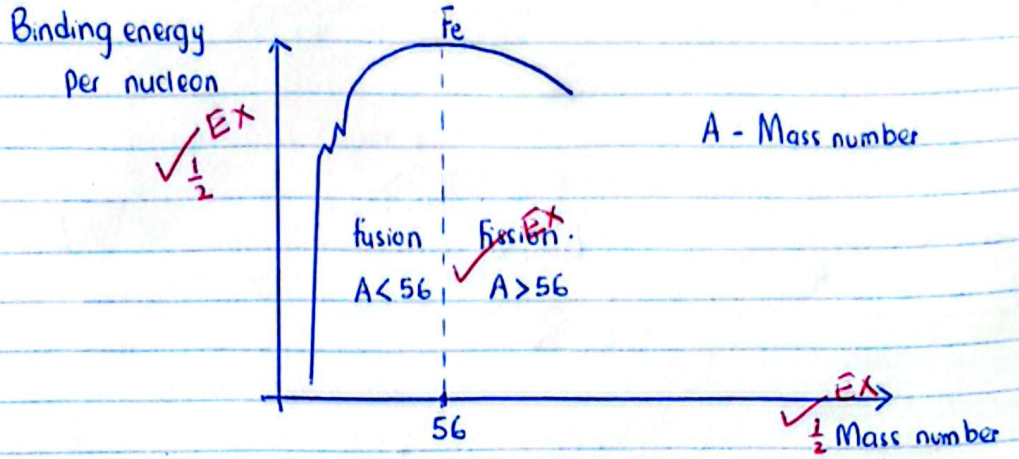
PHYSICS P510/1 SCORE GUIDE
BUGANDA MOCK EXAMINATIONS 2026 ①

ITEM 1

Part a Concerns of health Workers:

1) How the Binding Energy per nucleon graph explain atomic stability.

A graph of Binding Energy per nucleon against Mass number



- The higher the Binding energy per nucleon the more stable the atoms are.
- Low mass number elements have low Binding energy per nucleon but are generally stable because their proton to neutron ratio is almost 1
- The spikes represent abnormally stable atoms whose proton to neutron ratio = 1 ie Helium, Bellyrium, Carbon and Oxygen and require more energy to break than their immediate neighbours.
- As mass number increases the Binding per nucleon increases till A = 56 where BE/N peaks. This represents the most stable element, Iron Fe-56
- for that reason, low Mass number elements always undergo nuclear fusion
- The BE/N decreases for heavier elements indicating less stability and higher chances of undergoing nuclear fission

2) From the decay law which states that the rate of disintegration of an unstable element is directly proportional to the number of atoms present at that time.

$\Rightarrow \frac{dN}{dt} \propto N$ where N - Number of present particles.

$\frac{dN}{dt} = -\lambda N$ where λ - decay constant ✓ AN

$\frac{dN}{N} = -\lambda dt$

$\int \frac{1}{N} dN = -\lambda \int dt$ ✓ AN

At $t = 0$, $N = N_0$

$t = t$, $N = N$

$\int_{N_0}^{N} \frac{1}{N} dN = -\lambda \int_0^t dt$

$\ln N \Big|_{N_0}^{N} = -\lambda t \Big|_0^t$ ✓ AN

$\ln N_2 - \ln N_0 = -\lambda t$

$\ln N/N_0 = -\lambda t$

$\boxed{N = N_0 e^{-\lambda t}}$ ✓ AN - This is the decay equation.

where $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$ where $t_{\frac{1}{2}}$ is Half life of the element

3) The rate of disintegration is the Activity (A) of the sample

$$A = \frac{dN}{dt} = \lambda N$$

but $N = \left(\frac{m}{M}\right) N_A$ where $M = 60g$

$N_A = 6.023 \times 10^{23}$ atoms.

$m = 30g$

$$N = \left(\frac{30}{60}\right) \times 6.023 \times 10^{23} \checkmark AN$$

$$N = 3.011 \times 10^{23} \text{ atoms} \checkmark AN$$

$$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}} = \frac{\ln 2}{5.27} \checkmark AN$$

$$= 0.1315 \text{ year}^{-1}$$

$$= 4.171 \times 10^{-9} s^{-1} \checkmark AN$$

$$A = (4.171 \times 10^{-9} \times 3.011 \times 10^{23}) \checkmark AN$$

$$A = 1.256 \times 10^{15} Bq \checkmark AN$$

The decay equation : $^{60}_{27}Co \longrightarrow {}^{60}_{28}Ni + {}^{0}_{-1}e + \gamma$ ✓ AN

Mass of Reactants = 59.93381U

Mass of Products = (59.93079 + 0.0005)

= 59.93129U

Mass defect = (59.93381 − 59.93129)U

= 0.00252U ✓ AN

Energy Equivalence of Mass defect, E = (0.00252 × 931)

= 2.34612 MeV ✓ AN

Energy from 3.011×10^{23} atoms = (2.34612 × 3.011×10^{23}) ✓ AN

= 7.064 MeV ✓ AN

The energy emission from 30g of Co-60 is 7.064 MeV which is above the threshold of 5.5MeV but the ✓ AN disintegration Rate is 1.256×10^{15} Bq which is above 9×10^{14} Bq so the Machine is not efficient

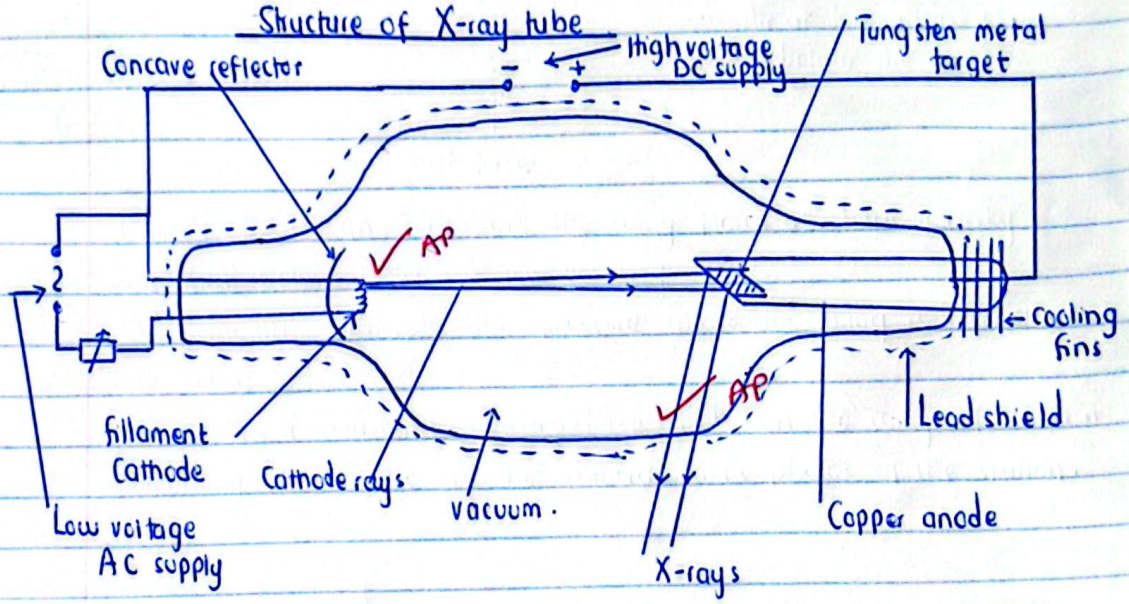
ITEM 1
Part B

The particles produced are beta particles. They have the following properties

- They are negatively charged ✓ AP
- They ionise gas atoms. ✓ AP
- They darken the photographic plate ✓ AP
- They are deflected by magnetic and electric fields ✓
- They have moderate penetrating power less than gamma rays and higher than alpha particles. ✓ AP

Any 2 properties.

-The particles with similar properties like these above are ✓ cathode rays AP
- Cathode rays are used in X-ray production ✓ AP in an X-ray tube



-The cathode is heated using a low voltage supply to emit electrons thermionically. ✓ AP
- The electrons are accelerated using high voltage to strike a high melting point target ~~embedded~~ ✓ AP in a copper anode and inclined at 45° to direction of electrons.
- About 99% of the ~~kinetic~~ ✓ AP Energy of the electrons is converted to heat energy which is driven away through the copper anode to the cooling fins to regulate heat and anode and ensure continuous and ✓ AP efficient operation of the tube

- About 1% of the Kinetic energy of the cathode rays is converted to X-rays which are directed to the target part (fracture)

ITEM 2

Concern 1

a) (i) Height of Parking

In the Parking orbit, the Gravitational force on the satellite creates centripetal force.

$$\frac{GMm_s}{r^2} = \frac{m_s v^2}{r}$$ ✓AN

$$\Rightarrow v^2 = \frac{GM}{r}$$

but $v = \frac{2\pi r}{T}$

$$\Rightarrow GMT^2 = 4\pi^2 r^3$$

$$r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$ ✓AN

where T = 24 hours

$= (24 \times 3600)\,s$

$= 86400\,s$ ✓AN

since $r = R_e + h \Rightarrow h = r - R_e$

$$h = \sqrt[3]{\frac{GMT^2}{4\pi^2}} - R_e$$ ✓AN

Considering the information given, it was insufficient for the learners to obtain r or h

for a learner that used $G = 6.67 \times 10^{-11}$ and $R_e = 6.4 \times 10^6 m$

$M_e = 5.97 \times 10^{24} kg$

$$r = \sqrt[3]{\frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times (24 \times 3600)^2}{4\pi^2}}$$ ✓AN ✓AN

$r = 42164244m = 4.22 \times 10^7 m$

$h = 4.22 \times 10^7 - 6.4 \times 10^6 = 3.58 \times 10^7 m$ ✓AN

The Parking Height, $h = 3.58 \times 10^7 m$. ✓AN

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Concern 2. (Mechanical Energy of the satellite)

Mechanical Energy, $E = P.E + K.E$

$$ME = -\frac{GM_e m_s}{2R}$$

$$= -\frac{6.67\times10^{-11}(250)(5.97\times10^{24})}{2(4.22\times10^{7})}$$ ✓AN

$$= -1.179\times10^{9}\,J$$ ✓AN ✓AN

The mechanical energy of the satellite is $-1.179\times10^{9}\,J$

Concern 3 (Mechanical Energy decrease effect on satellite)

When the Mechanical Energy of the satellite reduces, the radius of its orbit reduces ~~and~~ ie drops to lower orbit ✓EX

The satellite looses gravitational Potential energy ✓EX leading to increase in Kinetic energy and its orbital velocity ✓EX increases.

The viscous drag / air resistance on the satellite ✓EX increases leading to production of alot of heat that burns it up.

ITEM 2

Concern 2

Using deceleration journey by friction force

$F_r = \mu R = (0.025 \times 1000 \times 9.81)$ ✓ AN

$F_r = 245.25N$ ✓ AN ✓ AN

from $F = ma \Rightarrow 1000a = 245.25$ AN

$a = -0.24525 ms^{-2}$ ✓ AN

Using $v^2 = u^2 + 2as$

$0^2 = u^2 + 2(-0.24525)(200)$ ✓ AN

$u = 9.9 ms^{-1}$ ✓ AN

Using $v = u + at \Rightarrow t_1 = \frac{-9.9}{-0.24525}$ ✓ AN

$t_1 = 40.367s$ ✓ AN

Using Net force, $F = P - F_r$ where P - Tractive force

$\Rightarrow ma = (1800 - 245.25) = 1554.75$

$a = \frac{1554.75}{1000}$ ✓ AN

$= 1.55475 ms^{-2}$ ✓ AN

Using $v^2 = u^2 + 2as$

$S = \frac{9.9^2}{2(1.55475)}$ AN $= 31.52m.$ ✓ AN

time t_2 for acceleration $= \frac{v}{a} = \frac{9.9}{1.55475}$ ✓ AN

$= 6.368s$ ✓ AN

Remaining Distance to be covered with uniform velocity of 9.9m/s

$S = (600 - 31.52)$

$= 568.483m.$ ✓ AN

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time, t_3 for uniform velocity, $t_3 = \frac{s}{u}$

$= \frac{568.483}{9.9}$ ✓ AN

$= 57.4225s$ ✓ AN

Total time of Journey, $T = t_1 + t_2 + t_3$

$T = (40.367 + 6.368 + 57.4225)$ ✓ AN

$T = 104.1805s$ ✓ AN

$T = 1.74$ minutes. ✓ AN.

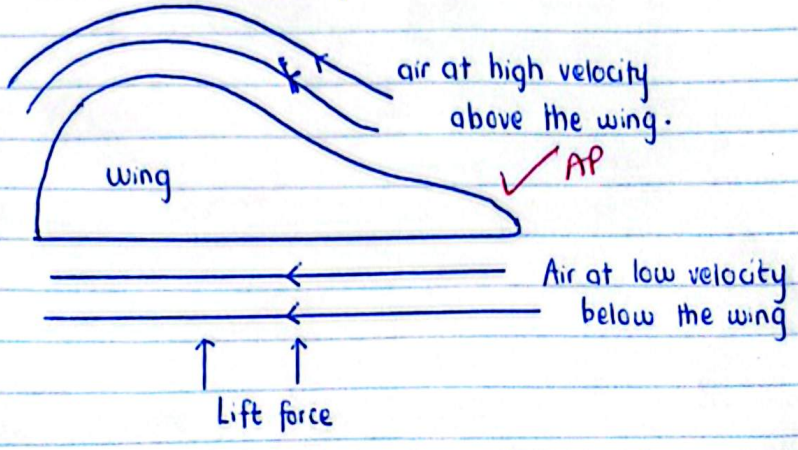
Since 1.74 minutes is the total time he spends in the journey and it's less than 5 minutes so he ✓ reported AN on time at the station.

ITEM 2 Part b

The device is a drone on which a camera is mounted.

- This will be designed in such a way that its wings are shaped as an aerofoil ✓AP
- The body will be streamlined to ensure minimal viscous drag of on the drone
-

Shape of the wing.



✓AP

Due to its design, air above the wing flows at high velocity and that below flows at low velocity ✓AP

By Bernoulli's principle, the pressure above the wing is lower than that below it ✓AP

This creates a resultant upward ~~force~~ (lift) on the wings of the dron mantaining it in space as the ✓ camera takes photos of the station.

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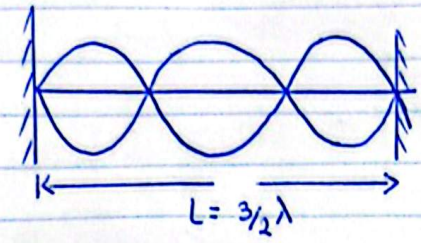
ITEM 3 (PART A)

Concern one (Tension in the stretched string / Guitar string)

a). $l = 40cm = 0.400m$

$m = 0.0012kg$

$f_2 = 750Hz$.



$\Rightarrow \lambda = \frac{2}{3}L$

$\lambda = \frac{2}{3}(0.4)$ ✓AN

$\lambda = 0.2667m$ ✓AN

$V = f\lambda$ from the wave equation

$V = (750)(0.2667)$ ✓AN

$V = 200ms^{-1}$ ✓AN ✓AN

Using $V = \sqrt{T/\mu}$ where $\mu = \frac{m}{L}$

$\Rightarrow T = \frac{V^2 m}{L}$

$\Rightarrow T = \frac{200^2(0.0012)}{0.4}$ ✓AN

✓AN

$T = 120N$ ✓AN

The tension in the stretched guitar string is 120N

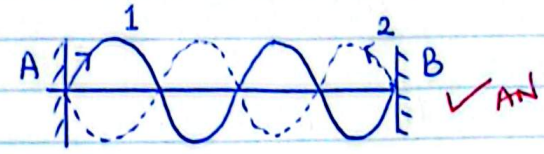
Concern two (2) The heating effect in the strings.

- The energy stored in the stretched string is $E = \frac{1}{2}Kx^2$ where K is the force constant and x is extension ✓ EX²

- As the string is released, some of the energy is used to cause vibration of air particles around the string so as to produce sound and the rest of the energy is converted to heat in the string eventually making them (strings) hot ✓ EX

Concern three (3) Derivation of equation of a stationary wave

Consider a string fixed at both ends A and B plucked at one end as below.



✓ AN

Wave 1 is a progressive wave travelling from A to B while wave 2 travels just opposite to 1 but with the same amplitude, frequency and velocity. The displacement of 1 and 2 at any time t are y_1 and y_2

$y_1 = A\sin(\omega t - kx)$ ✓ AN where K - wave number and A

$y_2 = A\sin(\omega t + kx)$ ✓ AN is Amplitude

By the Principle of superposition, of waves, when wave 1 and 2 superpose, $y = y_1 + y_2$ which is the displacement of the stationary wave

$\Rightarrow y = A[\sin(\omega t - kx) + \sin(\omega t + kx)]$ ✓ AN

from $\sin C + \sin D = 2\sin\left[\frac{C+D}{2}\right]\cos\left[\frac{C-D}{2}\right]$ ✓ AN

If $C = \omega t + kx$ and $D = \omega t - kx$

$\Rightarrow \frac{C+D}{2} = \omega t$ and $\frac{C-D}{2} = kx$

$\Rightarrow y = 2A\sin(\omega t)(\cos kx)$ AN

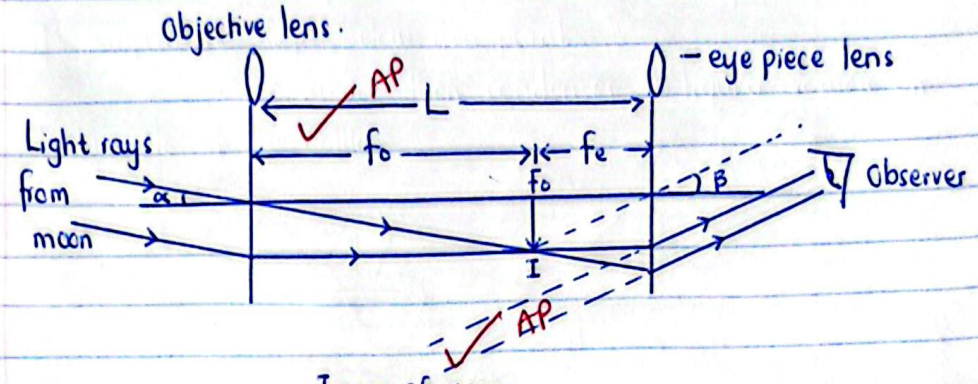
So $y = (2A\cos kx)\sin\omega t$ ✓ where $2A\cos kx$ is Amplitude of the wave

Concern 2 (Refracting Instrument)

- The refracting Instrument set under normal adjustment is the refracting telescope.

- The telescope is set in a way that the Image is formed at Infinity to allow the relaxed eye observe the moon with no strain.

Structure and Operation.



Design

- The objective lens has a wide aperture and a focal length fo greater than focal length, fe of the eye piece lens
- The two lenses are set a distance L = fo + fe apart

Mode of Operation.

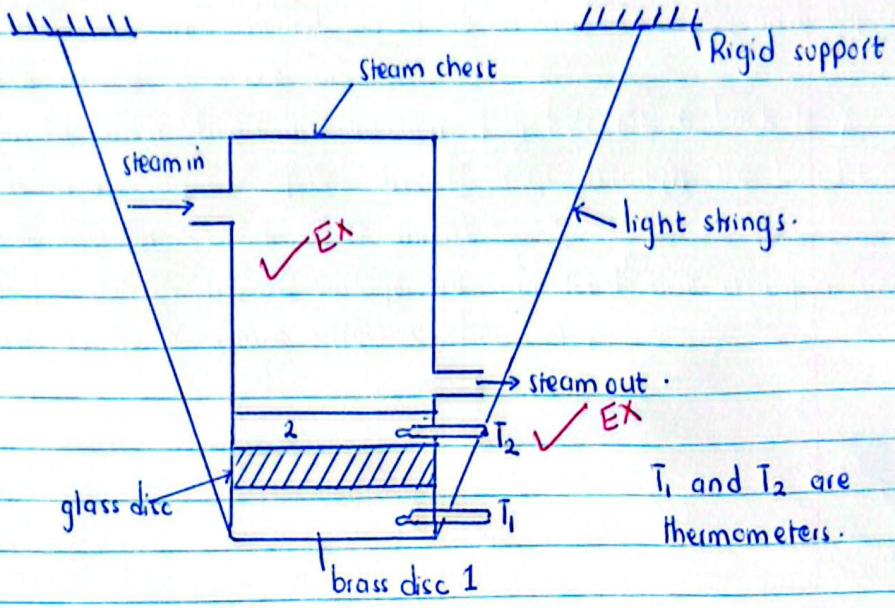
- The light rays from the moon enter the objective lens as a parallel beam at a small angle α
- The lens forms a real, diminished inverted, Image I at its focus Fo
- Since the eye piece lens is positioned in such away that its focus, fe coincides with fo, the Image I acts as an object at focus for the eye piece.
- The light rays from the object refract through the eye piece and emerge as a parallel beam at angle β forming a highly magnified, virtual Image at Infinity (Moon) as viewed by a relaxed eye
- The Magnification M of the Moon, $= \frac{f_o}{f_e}$

ITEM 4

Concern 1 (The thermal conductivity K_g of glass material)

Lee's Apparatus will be used since glass is a poor conductor

Experimental set up



- The glass is sliced into a circular thin disc of crossectional area, A and uniform thickness, x
- It is sandwiched between two brass discs 1 and 2 (attached to steam chest), each of mass, M and specific heat capacity, C in which holes are drilled and some mercury added for good thermal contact
- Thermometers T_1 and T_2 are fitted in the holes and steam passed through the chest continuously until T_1 and T_2 give steady readings. θ_1 and θ_2 respectively
- The glass disc is removed and steam chest chamber heats the disc 1 directly until its temperature is about 10°c above θ_1
- The chest is with drawn and glass disc placed back on the disc 1 and allowed to cool
- The temperature of disc 1 is recorded in equal time intervals until its temperature is about 10°c below θ_1

- A cooling curve is plotted and gradient, $\frac{d\theta}{dt}$ which is obtained from slope of the tangent drawn at θ is established
- From the Heat flow through glass disc = Heat loss rate through disc!

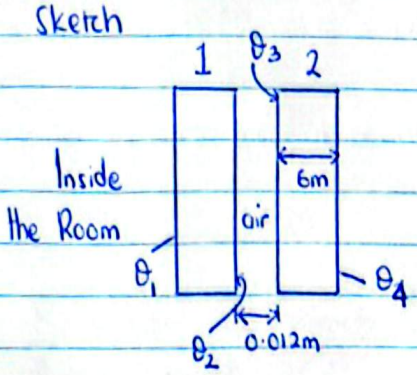
$$\Rightarrow \frac{K_g A(\theta_2 - \theta_1)}{x} = Mc\frac{(d\theta)}{dt}$$ EX

where K_g, the thermal conductivity of glass is calculated. EX

ITEM 4

Concern 2 (Could the classroom Inside temperature be maintained) at 20°c or below?

Sketch



Heat Flow Rate, $P = KA\frac{\Delta\theta}{L}$

$P_1 = P_2 = P_3$

$\Rightarrow P_1 = \frac{K_g A_g (\theta_2 - \theta_1)}{x_g}$

$\Rightarrow (\theta_2 - \theta_1) = \frac{P_1 x_g}{K_g A_g}$ —— ① ✓AN

$\Rightarrow P_2 = \frac{K_a A_a (\theta_3 - \theta_2)}{x_a}$

$\Rightarrow (\theta_3 - \theta_2) = \frac{P_2 (x_a)}{K_a A_a}$ —— ② ✓AN

$\Rightarrow P_3 = \frac{K_g A_g (\theta_4 - \theta_3)}{x_g}$

$\Rightarrow (\theta_4 - \theta_3) = \frac{P_3 (x_g)}{K_g A_g}$ —— ③ ✓AN

Adding equation ① + ② + ③

$\Rightarrow (\theta_2 - \theta_1) + (\theta_3 - \theta_2) + (\theta_4 - \theta_3) = \frac{P_1 x_g}{K_g A_g} + \frac{P_2 (x_a)}{K_a A_a} + \frac{P_3 x_g}{K_g A_g}$ ✓AN

$$\Rightarrow (\theta_4 - \theta_1) = \frac{P}{A}\left(\frac{2x_g}{K_g} + \frac{x_a}{K_a}\right)$$

Obtaining $\frac{2x_g}{K_g} + \frac{x_a}{K_a} = \left(\frac{2(6)}{0.8} + \frac{0.012}{0.025}\right)$ ✓AN

$= (15 + 0.48)$

$= 15.48\ m^2 K W^{-1}$ ✓AN

$\frac{P}{A} = \frac{(36 - \theta_1)}{15.48}$ which is Heat flow rate per m^2 ✓AN

but from Heat absorbed by window per second per m^2 $= \left(\frac{65}{100} \times 1400\right)$

$= 910\ W m^{-2}$ ✓AN

$\Rightarrow 910 = \frac{36 - \theta_1}{15.48}$ ✓AN

$\Rightarrow \theta_1 = -14050.8°C$ ✓AN ✓AN < Absolute zero (-273°C)

Basing on the value of Inside temperature which is -14050.8°C the classroom could be ~~maintained~~ AN by the windows below 20°C.

ITEM 4 Part B.

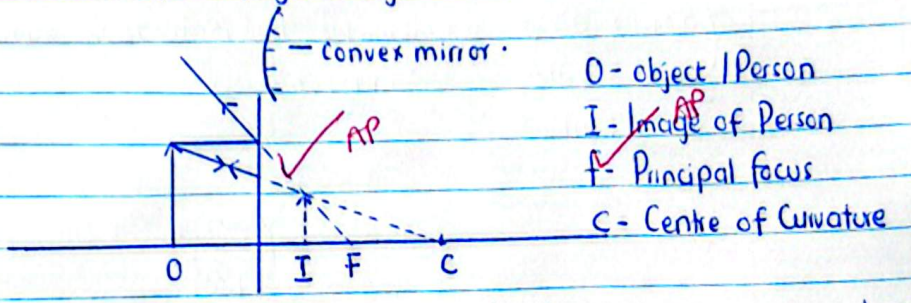
Modifications on the outer surface of the glass.

① Applying a highly reflective layer (tinting)
This will ensure high reflectivity and low heat absorption In the classroom thus keeping the room significantly cool.

The device Installed at the entrance is a Convex security mirror.

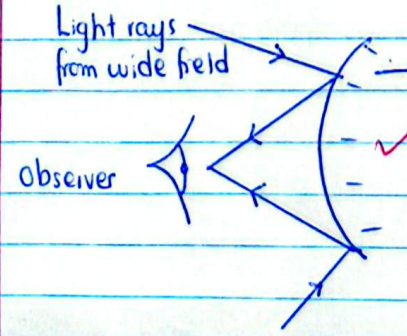
Reason: This provides a wide field of view of the whole area of entry and always gives upright Images of people trying to enter the room.

How it works to form upright Images.



- Paraxial light rays from the person are Incident on the mirror and are reflected off the mirror
- The rays are diverged and appear to come from focal point of the mirror.
- The light rays towards the centre of curvature are reflected back in the same path and appear to come from centre of curvature
- The two apparent rays intersect forming a virtual, upright diminished Image of the person approaching who is then monitored in his/her right orientation.

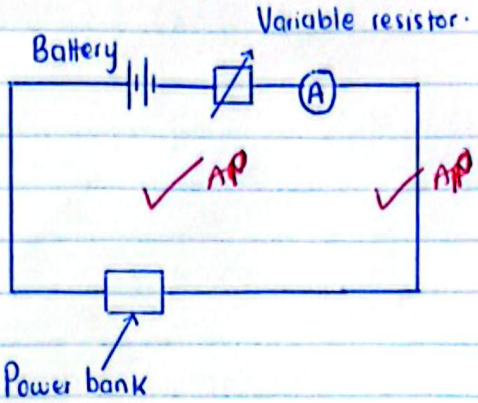
How it produces a wide field of view.



ITEM 5

Concern 1

Circuit Diagram by the S.5 student.



Emf and Internal resistance

Using $E = I(R+r)$

$E_1 = 0.48(21+r)$ —— ① ✓AN

$E_2 = 0.30(36+r)$ —— ② ✓

$\Rightarrow 0.48(21+r) = 0.30(36+r)$

$1.6(21+r) = 36+r$ ✓AN

$r = \left(\frac{2.4}{0.6}\right)$

$r = 4\Omega$ ✓AN ✓AN

$E = 0.48(21+4)$ ✓AN

$E = 12V$ ✓AN

The EMF of 12V is normal for standard charging but internal resistance of 4.0Ω is ✓AN significantly higher and cause a big voltage drop as current is being drawn, leading to a rapid shut down of the speaker.

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ITEM 5

Concern 2

Effective capacitance for series connection

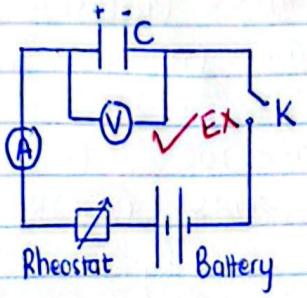
$$\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}$$ ✓AN

$$\frac{1}{C_{eff}} = \frac{n}{C} \Rightarrow C_{eff} = \frac{C}{n}$$ ✓AN

$$= \frac{1 \cdot 0}{n}$$ ✓AN

If the number of capacitors had been given then C_{eff} can be obtained

Concern 3 (Charging the capacitor)



- The capacitor plates are connected to a DC supply and switch, K closed. ✓EX
- This sets up an electric field between plates of the capacitor ✓EX
- The electrons are transferred from one plate to another creating positively charged and negatively charged plates. ✓EX
- This sets up an internal electric field in the capacitor that opposes that of the charging supply ✓EX
- The circuit voltage starts to drop and this results in a charge build up until the voltage across the capacitor is equal to the voltage of the supply ✓EX
- The current of the circuit falls to zero ie $I=0$ indicating full charging. ✓EX

ITEM 5

Concern 3 (Time for charging)

For maximum / Peak Power out put from a DC source during charging of a capacitor, $P_o = \frac{E^2}{4r}$ where E = Emf of DC source, r - Internal resistance

$P_o = \frac{12^2}{4(4)}$ ✓AN

$P_o = 9W$ ✓AN ✓AN

The energy stored in the capacitor system after full charge,

$E = \frac{1}{2}CV^2$

where V is the emf of the charging source at full charging.

$E = \frac{1}{2}(C_{eff})(12^2)$ ✓AN

$E = \frac{1}{2}(144)C_{eff}$

$E = 72\,C_{eff}$ in joules. $\Rightarrow E = \frac{72}{n}$ J ✓AN ✓AN

So from Energy, E = Power x Time

$\frac{72}{n} = 9t$ ✓AN

$\Rightarrow t = \frac{8}{n}$ s ✓AN ✓AN

The charging time, $t = 8/n$ s where n is number of capacitors. ✓AN

ITEM 6

Concern 1.

$$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$$

$$= \frac{4(8.85\times10^{-12})(5\times10^{-2})}{2\times10^{-3}}$$ ✓ AN

$$C = 8.85\times10^{-10}\,F$$ ✓ AN

Since they are in series arrangment; $C_r = \frac{1}{\left(\frac{1}{C_1}+\frac{1}{C_2}\right)}$

$$C_r = \frac{1}{(2/C)} = \frac{C}{2}$$ ✓ AN

$$C_r = \frac{8.85\times10^{-10}}{2}$$ ✓ AN

$$C_r = 4.425\times10^{-10}\,F$$ ✓ AN

Inductance, $L = \frac{\mu_0 (N^2) A_{ind}}{L}$

$$= \frac{[4\pi\times10^{-7}\times100^2\times1.2\times10^{-2}]}{[10\times10^{-2}]}$$ ✓ AN

$$= 1.5\times10^{-3}\,H$$ ✓ AN ✓ AN

At Resonance, Frequency, $f_o = \frac{1}{2\pi\sqrt{LC}}$

$$f_o = \frac{1}{2\pi\sqrt{(4.425\times10^{-10})(1.5\times10^{-3})}}$$ ✓ AN

$$f_o = 1.95\times10^{5}\,Hz$$ ✓ AN

$$f_o = 0.195\,MHz$$ ✓ AN

The Resonant frequency of the new circuit is 0.195MHz which is below the Operating frequency ✓ AN of the Radio channel (61.2MHz) so it can't tune to the desired religious ✓ AN channel unless the rator is adjusted to Increase capacitance so as to Increase resonant frequency to 61.2MH

ITEM 6

Concern 2 (How tuning of the circuit is done)

- Incoming Radio waves hit the aerial inducing an alternating current of the same high frequency AP in the primary aerial circuit.

- The A·C creates a changing magnetic flux which links with the tuning coil by mutual induction AP

- The changing magnetic flux induces an alternating emf in the inductor capacitor circuit AP

- The variable capacitor, C is adjusted until AP the resonant frequency fo matches the frequency of the religious station (61·2MHz)

- At resonance, the circuit produces maximum voltage at that specific frequency, selecting it out and unwanted AP frequencies are channelled to the ground (Earthed)

- After selection, the high frequency AC signal is passed through a diode which rectifies it to pulsating DC AP so that audio/sound can be extracted and sent to the speaker

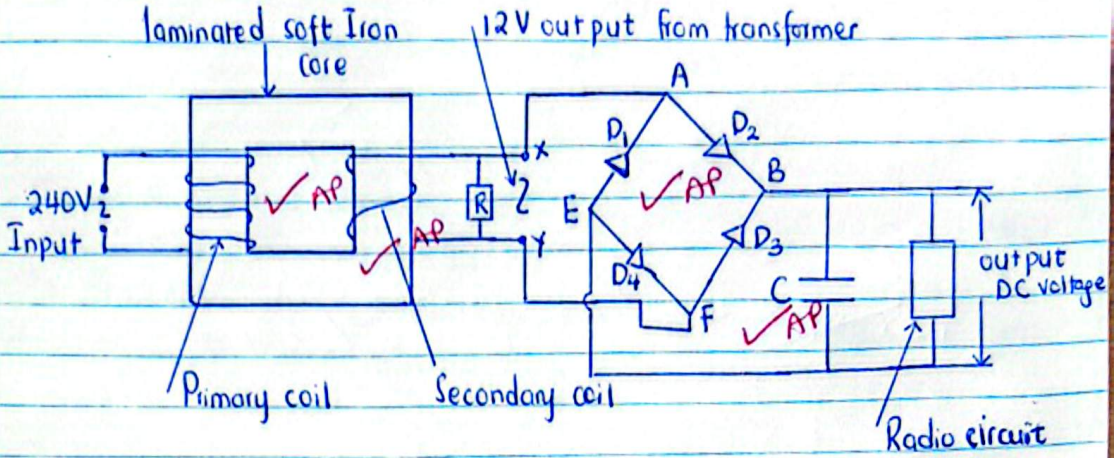
ITEM 6

Concern 3

Structure used by Radio charging device is made up of a step down transformer connected to a full wave rectification bridge circuit AP

- The transformer works on Mutual Induction and faraday's law of electromagnetism

Design of step down transformer, and rectification circuit AP



Operation of the Transformer.

- The 240V is fed in the Primary coil with more turns than secondary coil. AP
- An alternating current flows through the coil inducing a changing magnetic flux in the coil AP
- The changing Magnetic flux links to the secondary coil through the soft iron core by mutual Induction. AP
- The changing magnetic flux in the secondary coil generates an emf by faraday's law of electromagnetic Induction. AP
- The emf creates a low voltage out put in secondary coil across a load resistor, R due to a low turn ratio there. ie $\frac{N_p}{N_s} = \frac{V_p}{V_s} \Rightarrow N_p : N_s = 240 : 12 = 20 : 1$ AP

- The A·C output voltage of 12V is fed in a full wave Bridge rectifier circuit at X and Y terminals ✓ AP
- In the 1st half cycle, X is positive and Y is negative so diode D_1 and D_3 are forward biased so conduct while D_2 and D_4 are reverse biased so donot conduct current ✓ AP
- Current then flows in direction X to ABCDEF back to Y
- In the 2nd half cycle, Y is positive and X negative so diode D_1 and D_3 are forward biased so conduct while D_2 and D_4 are reverse biased so donot conduct. ✓ AP
- Current then flows in direction Y to FBCDEA back to X
- Thus in a complete cycle, current flows through the Load (Radio) in one direction CD ie direct current ✓ AP
- The fluctuations in DC voltage are smoothed out using a smoothing capacitor, C to ensure effective charging. ✓ AP